

Walter Andrade

Senior Software Engineer | Agentic AI, Voice Agents & Multi-Agent Systems | LangGraph, RAG, TypeScript, Python

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Summary

Senior Software Engineer with ~6 years building products end to end — specialized in agentic AI and LLM systems: voice agents, multi-agent orchestration with LangGraph (Python and JavaScript), Retrieval-Augmented Generation (RAG), function calling / tool use, structured outputs, and real-time streaming. Currently leading architecture and full-stack delivery of AI agent products for industrial knowledge management, including autonomous voice agents that handled 724 real interviews (55+ hours of conversation) across video meetings and telephony, and an AI agent workspace with retrieval and real-time chat. Sole DevOps owner on Azure, with LLM observability and cost tracking (~13% reduction in recurring cloud costs). Background in Philosophy (Logic), which shaped a rigorous, analytical approach to turning complex problems into clear, scalable systems. Experienced as tech lead, mentor, and product owner.

Experience

Senior Software Engineer

SmartHow | Belo Horizonte, Brazil (on-site)

09/2025 – Present

Lead architecture and full-stack delivery of AI agent products for industrial knowledge management within a 5-person engineering team — owning frontend, backend, AI layer, and DevOps on Azure.

- Led design and implementation of a **voice agent** platform for autonomous interviews: built the real-time voice pipeline in Python (FastAPI) — Whisper speech-to-text (STT), LangGraph agent orchestration, and low-latency ElevenLabs voice synthesis (TTS) over WebSocket streaming, handling turn-taking, interruptions, and end-to-end latency — plus Recall.ai meeting bots that autonomously join Google Meet, Zoom, and Teams. Voice agents handled 724 real interactions over ~9 months — 277 video-meeting interviews and 447 telephony calls, 55+ hours of autonomous conversation — replacing a fully manual interview operation.
- Architected and shipped a **multi-agent AI workspace** for industrial documentation (NotebookLM-style): LangGraph.js agent orchestration, real-time chat with asset ingestion — agents get direct tool access to uploaded documents, images, and external content — RAG retrieval (BM25), and structured outputs with human-in-the-loop approval (edit proposals). Cut content ingestion from a ~10-minute manual flow to seconds and fully automated image generation and insertion in documents.
- Designed the **agentic architecture** in Node.js (NestJS): tool-calling agents (function calling), async ingestion workers, and multi-agent workflows bridging the TypeScript backend and Python AI services.
- Reversed a churn-heavy delivery cycle — sprints were periodically consumed entirely by bug-fix work — by driving a full refactor of the core web application; boards now consistently hold a ~90% feature / 10% bug-fix split.
- Sole DevOps owner across CI/CD, testing, deployments, and Azure infrastructure; drove a ~13% reduction in recurring Azure costs via environment consolidation, Container App right-sizing, scale-to-zero, and LLM observability with per-feature cost attribution.
- Implemented observability and error tracking with Sentry, designed an AI-assisted code review workflow, and led sprint planning, ADRs, and cross-functional coordination across engineering and AI workstreams.

Skills: Agentic AI, Voice Agents, Multi-Agent Orchestration, LangGraph (Python & JS), RAG, Function Calling, Whisper, ElevenLabs, WebSockets, Node.js (NestJS), Next.js, Python (FastAPI), TypeScript, PostgreSQL, MongoDB, Azure, Sentry

Senior Full Stack Developer | Mentor | Product Owner

Instaq – Instituto Taqtile | Brazil (remote)

10/2023 – 09/2025

Led full-stack delivery across multiple client products, often acting as product owner and primary client interface.

- **Payssego** — Directed a Node.js payments platform for employee salary advances under strict API security requirements; built internal tools with Retool and ran secure onboarding flows that brought 1,100+ users live on day one of launch.
- **NaPorta** — Built a complex animated landing page (Next.js) and a fast admin system (Refine) for delivery operations management; the project won a national competition for technical quality, usability, and performance.
- **Pence** — Led product design, prototyping, and full-cycle development of a Culture Analytics Platform (React, Node.js, Neo4j), translating abstract culture data into actionable insights.
- **Modulart** — Led full-stack development of an events platform (modul.at.br) as Developer and Product Owner: responsive public website, PostgreSQL-backed backend, and comprehensive admin panel aligned with client operations.
- **Vivenda** — Developed an Astro platform connecting construction companies with people in vulnerable situations; owned the full AWS deployment, configuring an Ubuntu EC2 instance from scratch (networking, security groups, server setup).

Skills: Next.js, React, Node.js, TypeScript, PostgreSQL, Neo4j, Astro, Refine, Retool, AWS (EC2), GraphQL

Software Engineer

Taqtile Brasil | *São Paulo, Brazil*

09/2020 – 05/2023

- **Timesheet Platform** — Designed and built a full-stack Redmine replacement (React, GraphQL, Node.js, PostgreSQL) with BigQuery integration and Looker Studio dashboards for leadership; reduced operational costs and improved managerial decision-making.
- **Gentelab** — Led development of a full e-learning platform: React + Relay frontend, Hasura backend, Firebase authentication, video time tracking, and interactive assessments driving strong student engagement; the platform's codebase was later sold.
- **ioP Pet** — Developed a React Native app with Bluetooth integration, owning the full release process to the Apple App Store and Google Play.
- **Poli-USP Angel Platform** — Built a landing page that evolved into a lightweight social network for an angel investment group, growing the community from dozens to hundreds of active users.

Skills: React, React Native, GraphQL, Relay, Node.js, PostgreSQL, BigQuery, Hasura, Firebase

Open-Source Projects

Arachne — Rust TUI for visualizing git network graphs in the terminal.

Mr. Argus — Terminal process monitor for AI coding agents (TypeScript/Bun).

Education

BA in Philosophy (Philosophy of Mathematics | Logic) – Universidade Federal da Bahia, Brazil (2014 – 2019)

Full Stack Web Development – Labenu, Brazil (2020)

Languages

Portuguese: Native

English: Native / Bilingual proficiency

Spanish: Elementary

Skills

Agentic AI | AI Agents | Voice Agents | Multi-Agent Systems | LLM Orchestration (LangGraph — Python & JS, LangChain) | Retrieval-Augmented Generation (RAG) | Function Calling / Tool Use | Structured Outputs | Prompt Engineering | Speech-to-Text (Whisper) | Text-to-Speech (ElevenLabs) | Real-time Streaming (WebSockets) | LLM Observability & Cost Tracking | OpenAI & Anthropic APIs | TypeScript | JavaScript | Python | Go | Node.js | React | React Native | Next.js | GraphQL | REST APIs | PostgreSQL | MongoDB | Neo4j | Azure | AWS | GCP | Docker | CI/CD | Software Architecture